SOC Investigation Report

Network Traffic Analysis / Phishing Attack & C2 Communication

Analyst: Mohamed Hadjaz Tool: Wireshark + VirusTotal · Lab:CTF challenge

1. Incident overview

Incident type	Phishing / malicious email attachment
Victim endpoint	10.9.23.102
PCAP file	carnage.pcap
Alert trigger time	16:44:38
Malicious domains identified	4
Malicious IPs identified	5
Payload	documents.zip → chart-1530076591.xls
Verdict	True positive →confirmed phishing & C2 activity

An endpoint (10.9.23.102) generated a security alert after the user opened a document attached to a phishing email. The attachment triggered a download of a malicious ZIP archive, followed by a series of encrypted outbound connections to four external domains, consistent with a multi-stage phishing attack leading to active C2 communication. The investigation used Wireshark to analyse the PCAP file carnage.pcap, with threat intelligence verified via VirusTotal.

2. Attack timeline

Time	Event
16:44:38	User opens malicious email attachment
16:44:38	documents.zip downloaded from attirenepal.com (85.187.128.24)
16:45:07	Outbound TLS to 13.107.42.16 (Microsoft infrastructure, benign)
16:45:11	C2 beacon to finejewels.com.au (148.72.192.206)
16:45:21	C2 beacon to thietbiagt.com (210.245.90.247)
16:45:25	C2 beacon to new.americold.com (140.72.53.144)
17:00:03+	Sustained C2 beaconing from 23.111.114.52 (Russia, AS 39134)

3. Detailed investigation

3.1 Initial payload delivery: attirenepal.com

At 16:44:38, the endpoint sent an HTTP GET request to attirenepal.com (85.187.128.24), downloading documents.zip from the path /incidunt-consequatur/documents.zip.

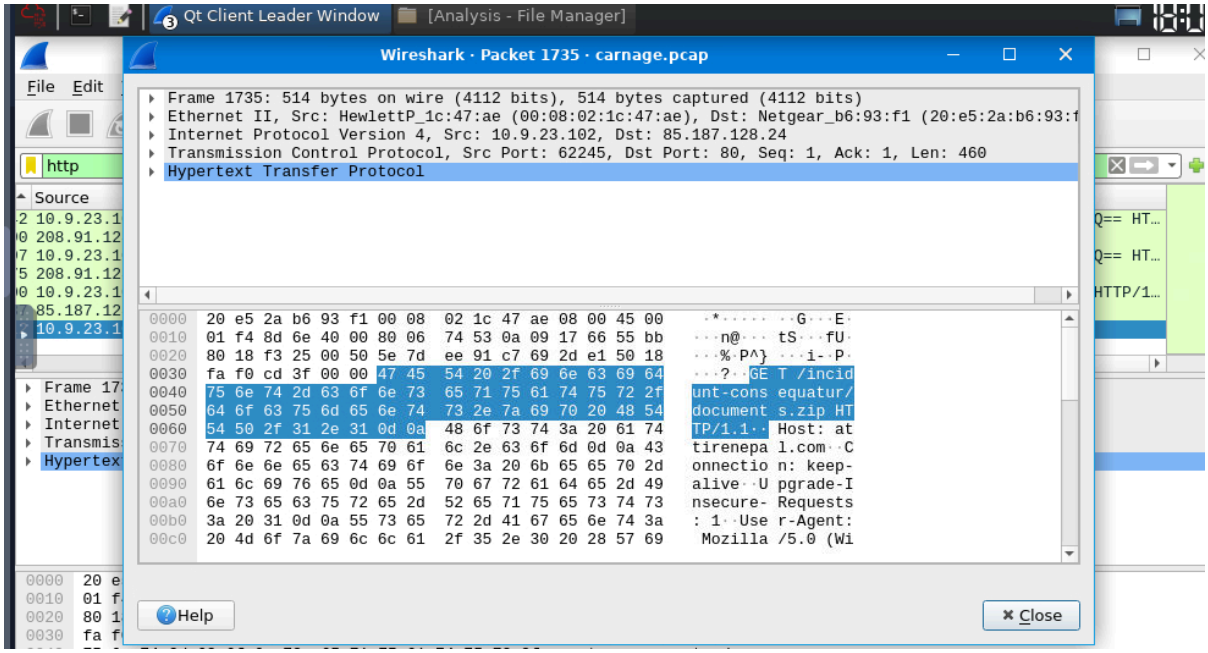

Figure 1: Packet 1735: HTTP GET request to attirenepal.com (85.187.128.24)

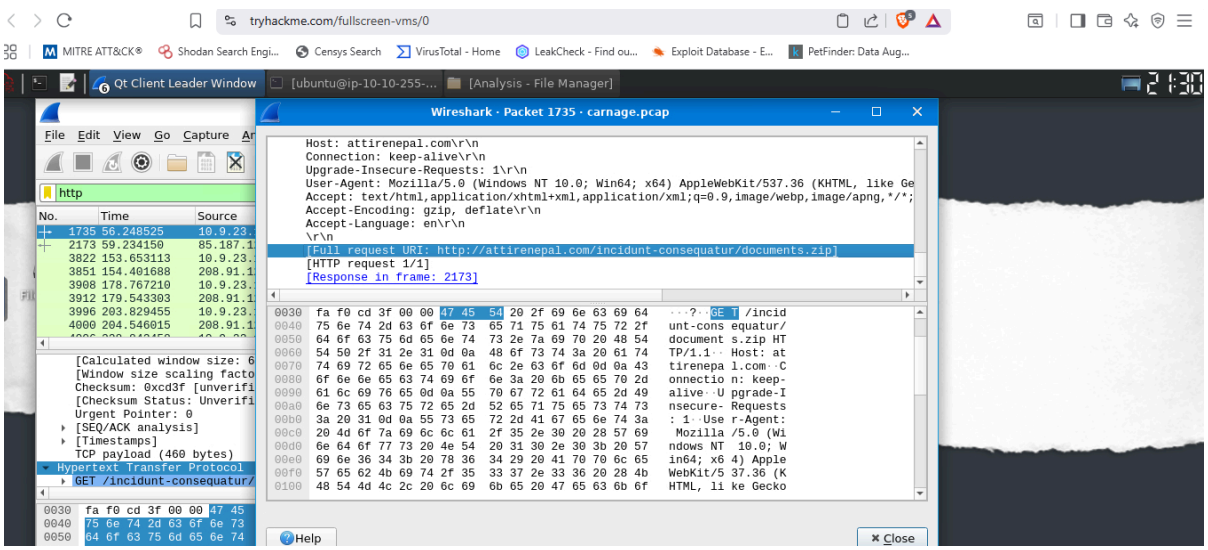

Figure 2: Full HTTP request confirming the documents.zip download URL

Following the HTTP stream revealed the server was running LiteSpeed with PHP/7.2.34. The file was served as an octet-stream attachment with HTTP 200 OK.

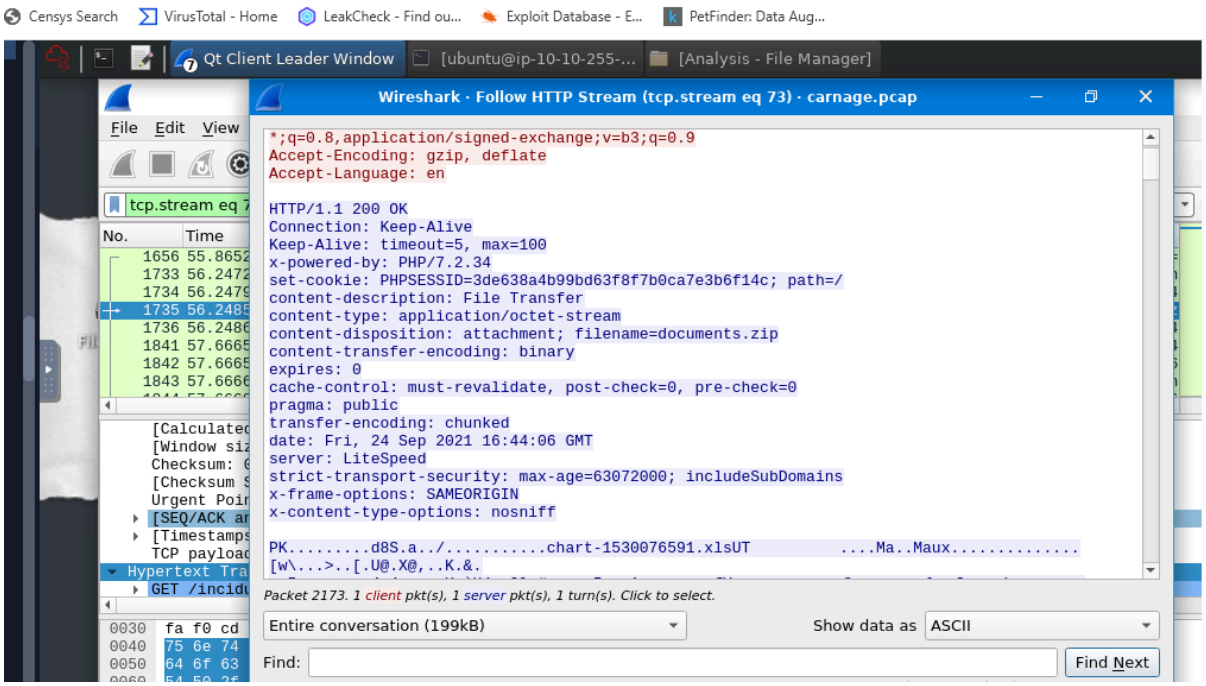

Figure 3: HTTP stream showing LiteSpeed/PHP 7.2.34 server headers and documents.zip response

VirusTotal flagged the URL as malicious (6/98 vendors), confirmed by CyRadar, ESET, Fortinet, and Kaspersky.

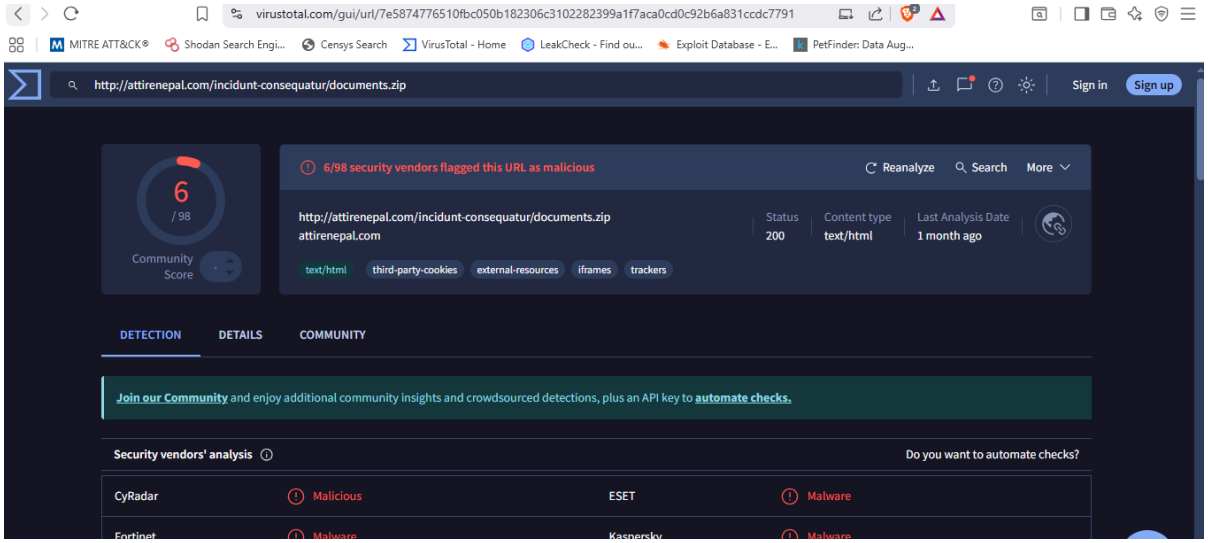

Figure 4: VirusTotal: attirenepal.com flagged malicious by 6/98 vendors

Sandbox extraction of documents.zip revealed a single file: chart-1530076591.xls, an XLS payload consistent with Excel macro-based malware.

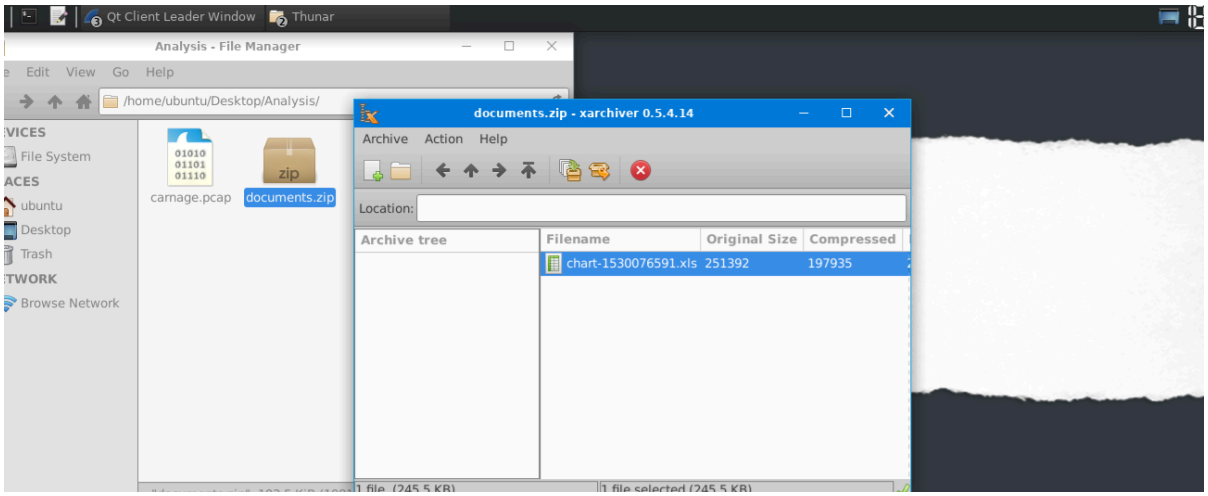

Figure 5: Sandbox extraction showing chart-1530076591.xls inside documents.zip

3.2 C2 beacon: finejewels.com.au (148.72.192.206)

At 16:45:11, the endpoint initiated a TLS connection to finejewels.com.au (148.72.192.206). The destination was confirmed via Client Hello inspection prior to encryption. VirusTotal returned a 4/98 malicious score.

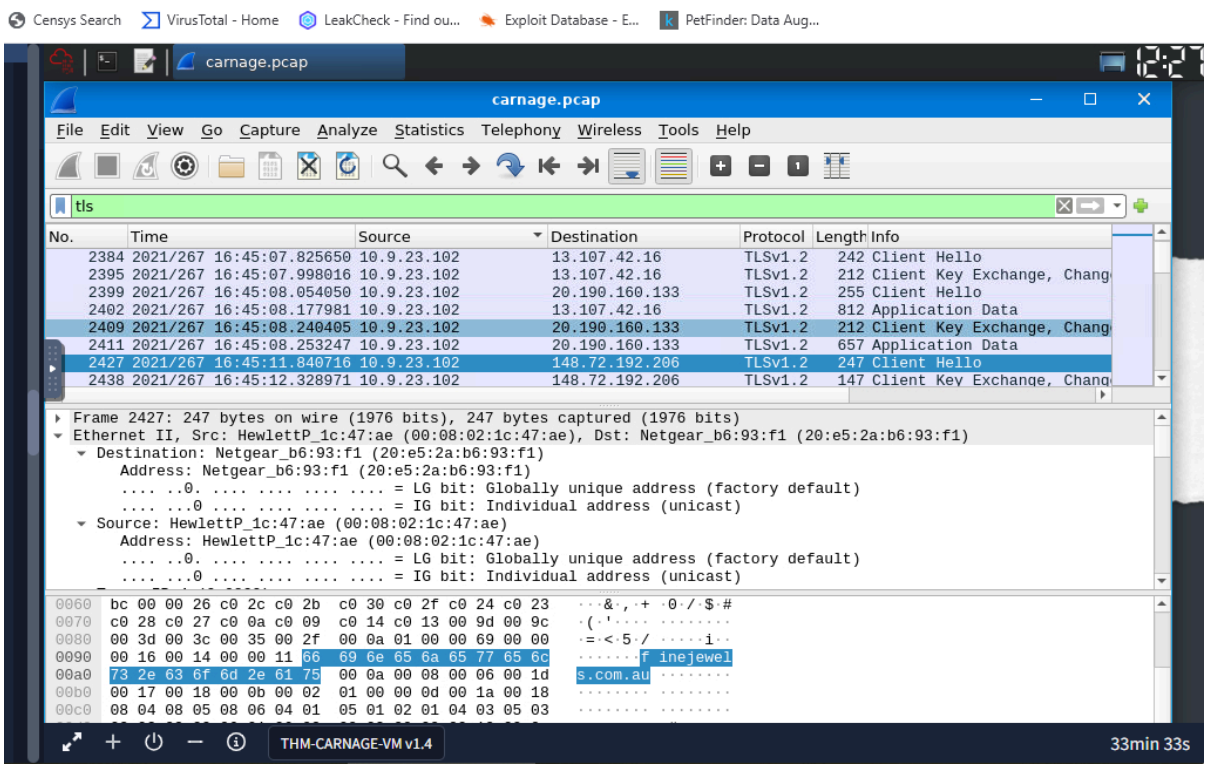

Figure 6: TLS filter showing Client Hello to finejewels.com.au at 16:45:11

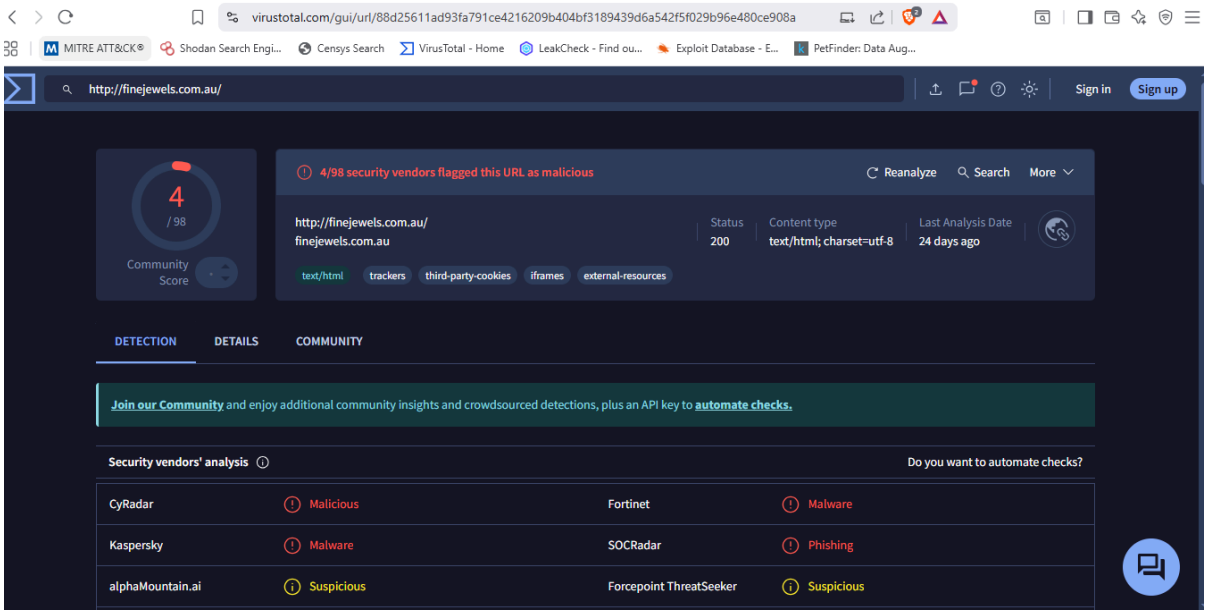

Figure 7: VirusTotal: finejewels.com.au flagged by 4/98 vendors

Following the TCP stream showed the SSL certificate was issued by GoDaddy, a common tactic where attackers register cheap domains with legitimate certificate authorities to bypass SSL-based filtering.

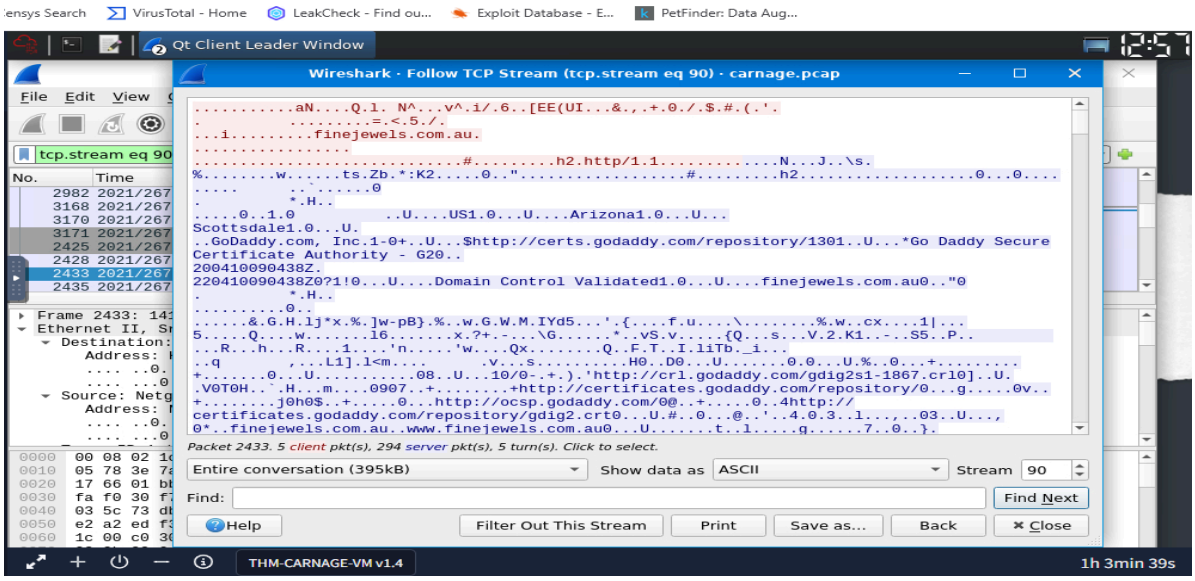

Figure 8: TCP stream revealing GoDaddy-issued SSL certificate for finejewels.com.au

3.3 C2 beacon: thietbiagt.com (210.245.90.247)

At 16:45:21, a further TLS connection was made to thietbiagt.com (210.245.90.247). VirusTotal flagged this domain as malicious (3/95), registered under MAT BAO CORPORATION.

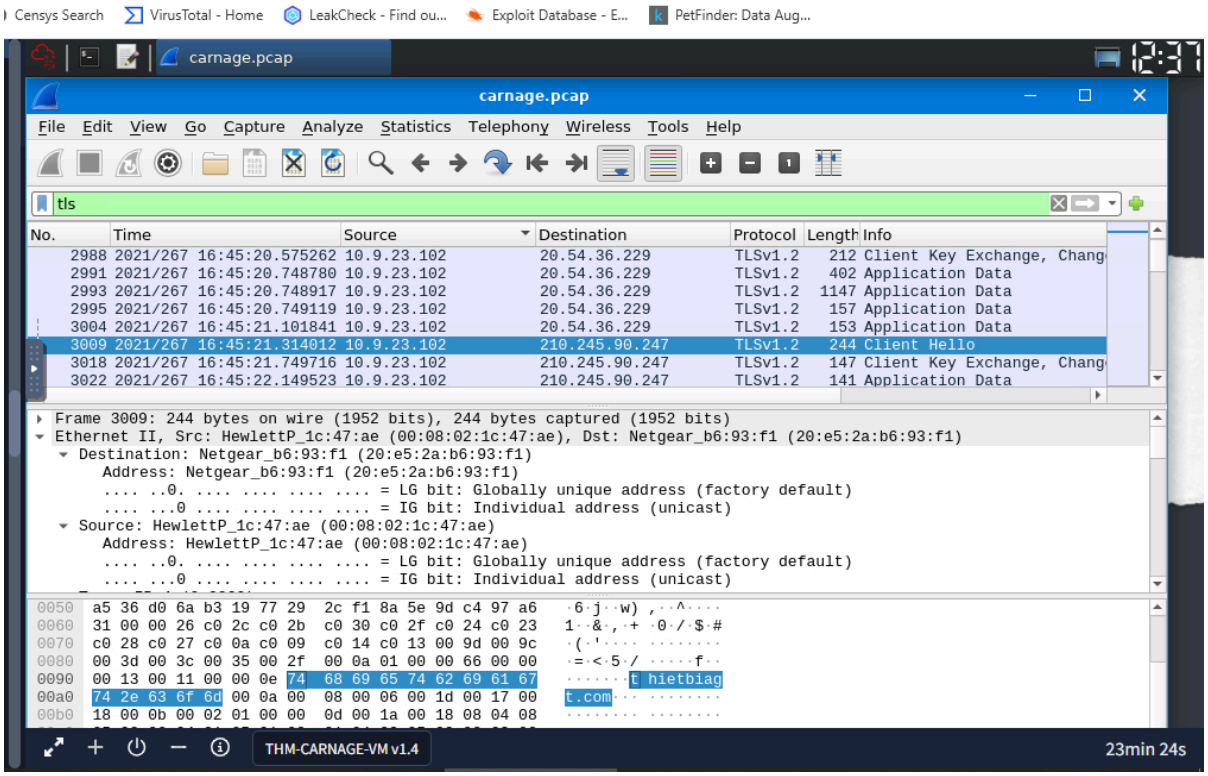

Figure 9: Wireshark: Client Hello to thietbiagt.com (210.245.90.247) at 16:45:21

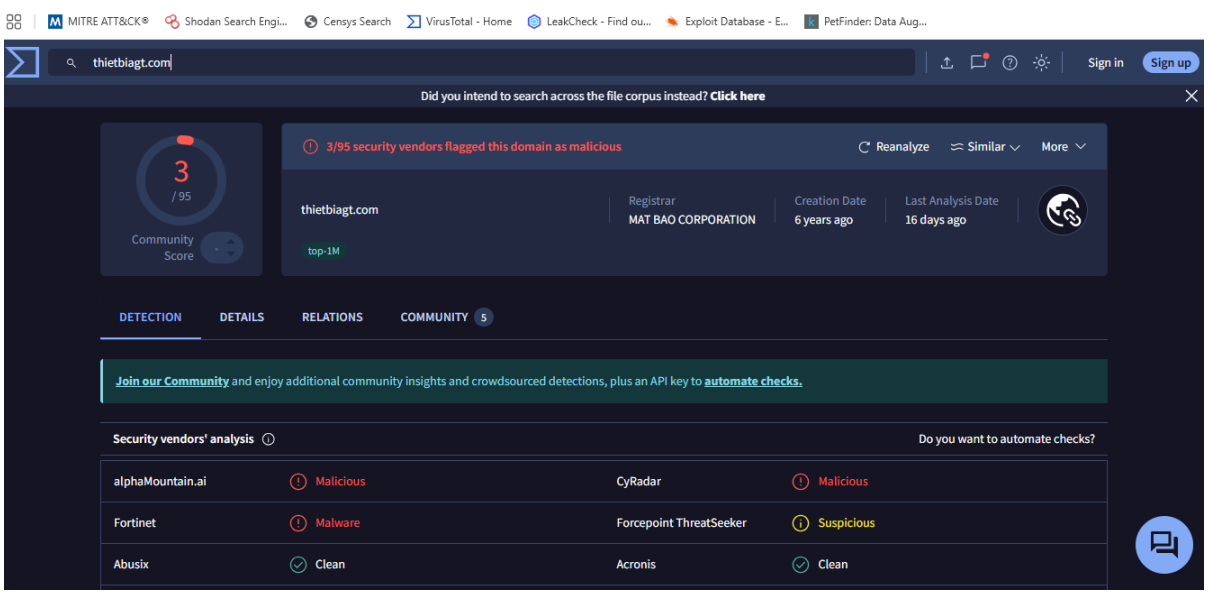

Figure 10: VirusTotal: thietbiagt.com flagged by 3/95 vendors

3.4 C2 beacon: new.americold.com (140.72.53.144)

At 16:45:25, the endpoint connected to new.americold.com (140.72.53.144). VirusTotal returned only 1/95 (Fortinet). Despite the low score, the connection follows the same TLS beacon pattern as the previous two confirmed C2 channels and should be treated as malicious pending further analysis.

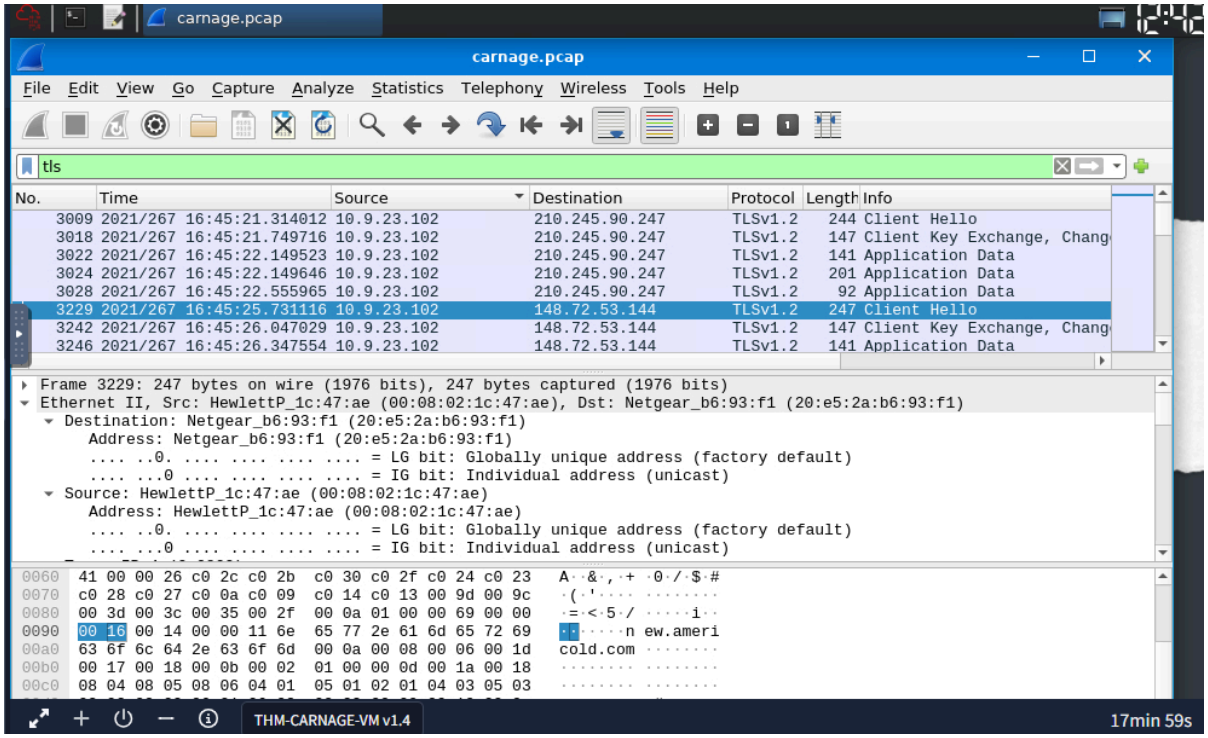

Figure 11: Wireshark: Client Hello to new.americold.com (140.72.53.144) at 16:45:25

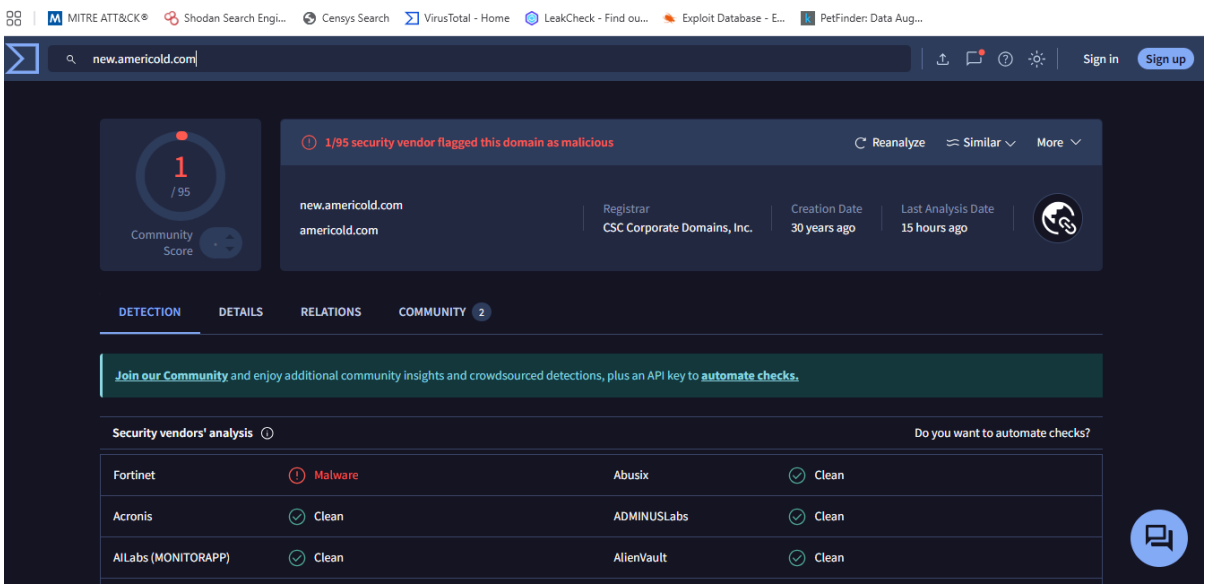

Figure 12: VirusTotal: new.americold.com (1/95 low detection, treat as suspicious)

3.5 Post-compromise activity — 23.111.114.52

From 17:00:03 onwards, sustained inbound TLS connections were observed from 23.111.114.52 (AS 39134, Edinaya Set Limited Liability Company, Russia). The repeated "Ignored Unknown Record" pattern is characteristic of an active C2 implant maintaining persistent communication. VirusTotal flagged this IP as malicious by 13/95 vendors including BitDefender, ESET, CyRadar, and alphaMountain.ai.

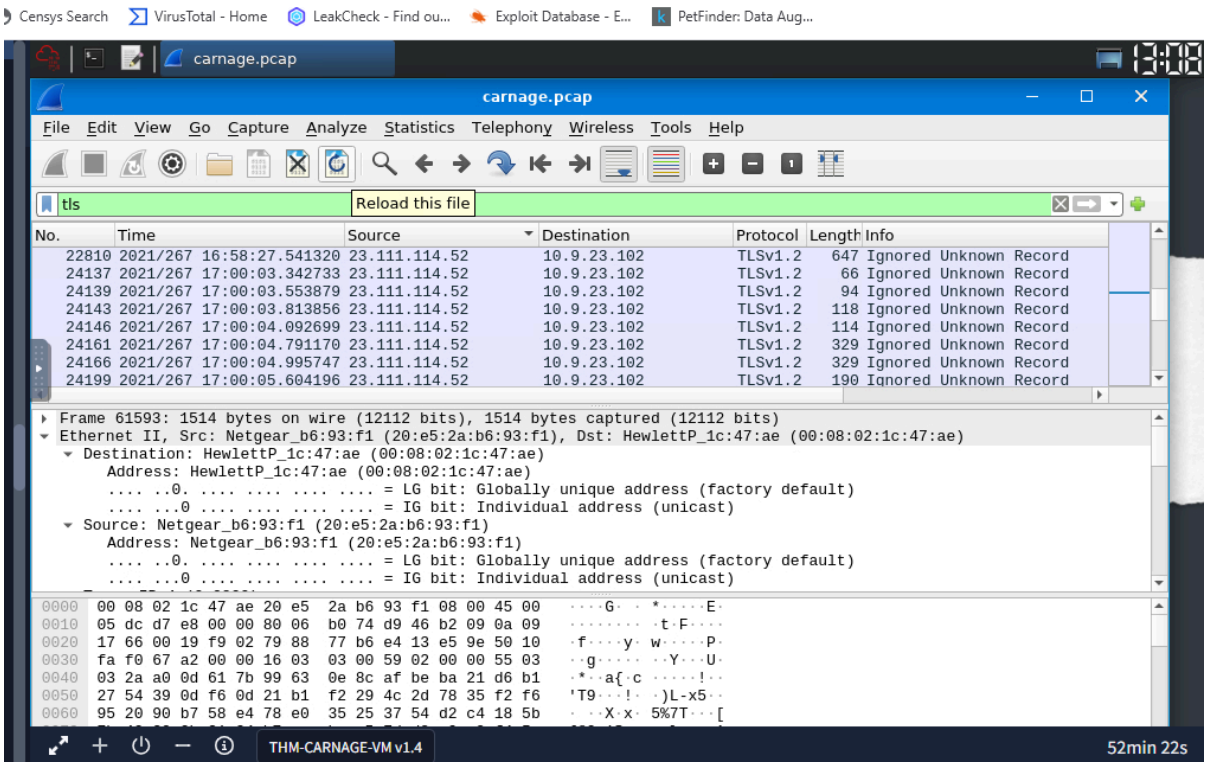

Figure 13: Sustained TLS beaconing from 23.111.114.52 to the victim endpoint from 17:00 onwards

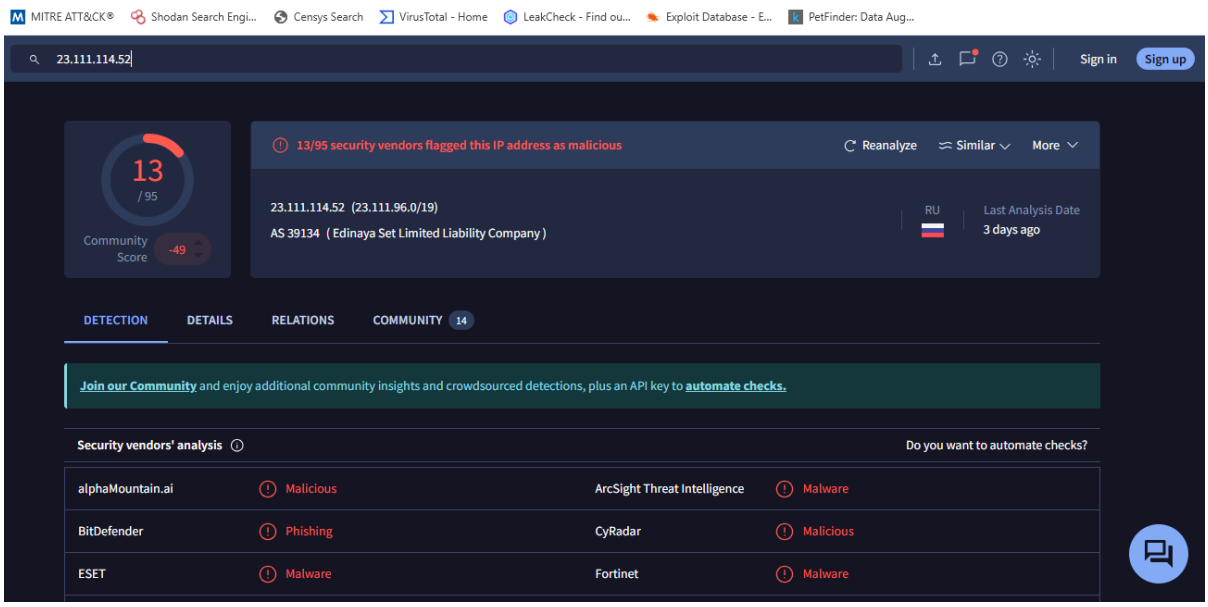

Figure 14: VirusTotal: 23.111.114.52 flagged by 13/95 vendors, Russia-based AS 39134

4. Indicators of compromise

Domain	IP address	First seen	VT score	Notes
attirenepal.com	85.187.128.24	16:44:38	6/98	Initial payload delivery
finejewels.com.au	148.72.192.206	16:45:11	4/98	C2 beacon, GoDaddy SSL cert
thietbiagt.com	210.245.90.247	16:45:21	3/95	C2 beacon, MAT BAO registrar
new.americold.com	140.72.53.144	16:45:25	1/95	C2 beacon, low detection
23.111.114.52	23.111.114.52	17:00:03+	13/95	Post-compromise C2 beaconing

5. MITRE ATT&CK mapping

ID	Technique	Tactic	Observed
T1566.001	Phishing: Spearphishing Attachment	Initial Access	Malicious XLS via email attachment
T1204.002	User Execution: Malicious File	Execution	User opened attachment, triggered payload
T1105	Ingress Tool Transfer	C2	documents.zip downloaded from C2 server
T1071.001	Application Layer Protocol: Web	C2	HTTP/TLS beaconing to 4 external domains
T1573.001	Encrypted Channel	C2	TLS-encrypted traffic to multiple C2s
T1082	System Information Discovery	Discovery	Post-compromise recon activity observed

6. Containment and recommendations

Severity and impact assessment

Severity: High

The compromise of endpoint 10.9.23.102 indicates a high risk to organizational security. The successful execution of a malicious XLS payload and subsequent C2 communication suggests the attacker may have gained remote access to the system.

Potential impacts include:

- Unauthorized data exfiltration
- Remote command execution via C2 channels
- Establishment of persistent access
- Possible lateral movement to other systems

7. Containment and recommendations

Immediate actions

- Isolate endpoint 10.9.23.102 from the network to prevent further C2 communication and lateral movement.
- Block all identified IPs and domains at the perimeter firewall and DNS resolver: 85.187.128.24, 148.72.192.206, 210.245.90.247, 140.72.53.144, 23.111.114.52.
- Escalate chart-1530076591.xls to a malware analyst for static and dynamic analysis.
- Perform a full memory and disk acquisition of the compromised endpoint before remediation.
- Hunt across SIEM/EDR logs for other endpoints that communicated with the identified IOCs.

Preventive measures

- Implement email gateway filtering to strip or quarantine macro-enabled Office attachments from external senders.
- Deploy DNS-based threat filtering to block connections to known malicious domains at the resolver level.
- Create detection rules alerting on rapid outbound TLS connections to multiple new domains shortly after a document is opened.
- Run phishing awareness training for end users, with emphasis on email attachments from unknown senders.

8. Conclusion

This investigation confirms a true positive phishing alert. The compromised endpoint received a malicious XLS payload via email, which established C2 channels to four external domains. Sustained post-compromise beaconing from a Russia-based IP (23.111.114.52) confirms the implant was actively communicating well beyond the initial infection window.

The methodology that I applied here: timestamp isolation, HTTP stream analysis, TLS Client Hello inspection, sandbox extraction, and VirusTotal validation, successfully identified all malicious connections. The same approach can be extended to remaining unanalysed traffic in the capture to surface any additional IOCs.